MIL-H-22995A(AS)
25 April 1969
Superseding
MIL-H-22995 (Wep)
27 July 1961

MILITARY SPECIFICATION

HELMET SHELL ASSEMBLIES, PILOT'S PROTECTIVE, APH-6 SERIES

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers the requirements for the pilot's protective APH-6 series helmet shell assembly.

1.2 Classification - The pilot's protective helmet shell assemblies shall be furnished in two sizes, medium and large.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

NN-P-530	Plywood, Flat Panel
UU-P-268	Paper, Kraft, Untreated, Wrapping
PPP-B-576	Box, Wood, Cleated, Veneer, Paper Overlaid
PPP-B-591	Box, Fiberboard, Wood-Cleated
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-C-00795	Cushioning Material, Cellular, Plastic Film, (for Packaging Application)

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SPECIFICATIONS (Continued)

Federal

PPP-T-60	Tape, Pressure-Sensitive Adhesive, Waterproof, for Packaging

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property

DRAWINGS

Aerospace Crew Equipment Department

32H1291-11	Helmet, Medium, Pilot's Protective, APH-6 Series
32H1292-11	Helmet, Large, Pilot's Protective, APH-6 Series

(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government quality assurance representative. Copies of this specification and other unclassified specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Publications and Forms Center, (Code 1051), 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.)

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3. REQUIREMENTS

3.1 Qualification - The helmet shell assemblies furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 First article - The helmet shell assemblies furnished under this specification shall be a product which has been inspected and passed the first article inspection specified herein (see 4.4).

3.3 Materials - Materials used in the construction of the helmet shell assemblies shall conform to the applicable specifications and shall be as specified herein and on the applicable drawings.

3.4 Construction - The helmet shell assemblies shall be constructed in accordance with ACEL Drawings 32H1291-11 and 32H1292-11.

3.4.1 Molding - The helmet shell assemblies shall be molded in such a manner as to produce no distortion or warpage. Wherever molding materials are cut to patterns prior to molding, pattern shapes shall be such as to produce maximum strength in the final shell.

3.4.2 Finish - The helmet shell assemblies shall be prepared for the finishing process by first cleanly punching and drilling all necessary holes at the required locations. All flash marks shall be removed and the external surfaces finished smooth. The surfaces to be finished shall then be coated with a paint system consisting of a surface primer and a white lacquer. The color and gloss of the finished helmet shell assembly shall be in accordance with color No. 17875 of FED-STD-595. After a minimum drying time of twelve hours, the paint finish shall be completely smooth to the finger tips and exhibit good adhesion when tested as specified in Section 4.

3.5 Performance -

3.5.1 Piercing resistance - The degree of penetration, when tested as specified in 4.6.2, shall not exceed 1/8 inch protrusion.

3.5.2 Impact - Any acceleration in excess of 400g or evidence of "bottoming", when tested as specified in 4.6.3, shall be cause for rejection.

3.5.3 Finish adhesion - When tested as specified in 4.6.4 any evidence of lifting of any of the squares shall be cause for rejection.

3.6 Workmanship - Workmanship shall be in accordance with high grade practices for the processes involved. The helmet shell assemblies shall be clean and well finished and shall completely meet the quality conformance levels as specified in Section 4.

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3.7 Identification of product - Each helmet shell assembly shall be identified in accordance with MIL-STD-130 and shall have a permanent type label applied to the exterior surface as shown on the applicable drawing and shall contain the following information:

HELMET SHELL ASSEMBLY, PILOT'S PROTECTIVE,
APH-6 SERIES
MIL-H-22995A(AS)
SIZE
CONTRACT NO.
CONTRACTOR'S NAME

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection - The inspection and testing of the helmet shell assemblies shall be classified as follows:

(a) Qualification inspection - Qualification inspection consists of tests and examinations performed on samples submitted for approval as qualified products (see 4.3).

(b) First article inspection - First article inspection consists of tests and examinations performed on helmet shell assemblies which are representative of the production item after award of a contract to determine that the production item conforms to the requirements of this specification (see 4.4).

(c) Quality conformance inspection - Quality conformance inspection consists of tests and examinations performed on individual products or lots to determine conformance of the products or lots with the requirements set forth in this specification (see 4.5).

4.3 Qualification inspection - Qualification inspection shall consist of all the tests and examinations listed in this specification.

4.3.1 Qualification inspection samples - The qualification inspection samples shall consist of three helmet shell assemblies. The sizes of the three samples shall be two medium and one large. The qualification samples shall be forwarded to the

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Supply Officer, Naval Air Development Center, Johnsville, Warminister, Pennsylvania 18974, Attention: Director, Aerospace Crew Equipment Department. The samples shall be identified with the following information:

HELMET SHELL ASSEMBLIES, PILOT'S PROTECTIVE, APH-6 SERIES
Name of Manufacturer
Manufacturer's designation or number
Submitted by (name) (date) for qualification inspection in accordance with the requirements of MIL-H-22995A(AS) under authorization (reference authorizing letter).

4.4 First article inspection - The first article inspection shall consist of the following tests and examinations:

Visual examination
Dimensional examination
Piercing resistance
Impact
Finish adhesion

4.4.1 First article samples - Unless otherwise specified, as soon as practicable after award of the contract or order, the manufacturer shall submit two helmet shells, one large and one medium. The samples shall be representative of the construction, workmanship, components and materials to be used during production. When a manufacturer is in continuous production of the helmet shell assemblies from contract to contract, submission of further first article inspection samples on the new contract may be waived at the discretion of the procuring activity (see 6.2). Approval of the first article inspection sample or waiving of the first article inspection does not preclude the requirements for performing the quality conformance inspection. The first article inspection samples shall be furnished the Government as directed by the contracting officer (see 6.2).

4.4.2 Upon the completion of the first article inspection, all the applicable inspection reports, and when applicable, recommendations and comments pertinent for use in monitoring production shall be forwarded to the cognizant Government activity. One helmet shell assembly shall be returned to the manufacturer for use in monitoring production. The other helmet shell assembly will be consummed or destroyed in the first article inspection and shall not be considered as part of the quantity to be delivered under contract.

4.5 Quality conformance inspection - The quality conformance inspection shall consist of the following;

Visual examination
Dimensional examination
Preparation for deliver

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4.5.1 Sampling -

4.5.1.1 Inspection lot - An inspection lot shall be expressed in units of helmet shell assemblies, made under essentially the same conditions and from the same materials and components. The sample unit shall be one helmet shell assembly.

4.5.1.1.1 Preparation for delivery - An inspection lot shall be expressed in units of one fully prepared shipping container, containing a helmet shell assembly, fully prepared for delivery, made from essentially the same materials and components. The sample unit shall be one shipping container, containing a helmet shell assembly, fully prepared for delivery with the exception that it need not be sealed.

4.5.1.2 Sampling for tests and examination of helmet shell assemblies - The sample size, acceptance criteria, tests and examinations required for the helmet shells shall be as specified in Table I.

4.6 Inspection methods -

4.6.1 Visual examination - Each helmet shell assembly selected in accordance with Table I shall be examined visually to determine conformance with this specification. The classification of defects, Table II, shall be used to classify the defects found. For dimensional defects, each helmet shell assembly selected as a sample unit from the lot shall be thoroughly checked dimensionally to determine conformance to this specification. The classification of defects, Table III, shall be used to classify the defects found.

4.6.2 Preparation for delivery - Each of the fully prepared shipping containers, containing helmet shell assemblies, selected as a sample unit from the lot shall be visually examined to determine that the packaging, packing, and marking conform to this specification. The classification of defects, Table IV, shall be used to classify the defects found.

4.6.3 Piercing resistance - The helmet shell assemblies shall be subjected to the impact of a 16-ounce steel plumbob, pointed to 60 degrees and dropped (free fall) from a height of 10 feet onto the outside surface of the helmet in a direction perpendicular to the surface. The points of impact shall be one in each 60-degree quadrant of the helmet at a radial distance 4-1/2 ±1/2 inches from the apex and also at the apex. The helmet shell assembly shall meet the requirement of 3.5.1.

4.6.4 Impact evaluation -

4.6.4.1 Impact evaluation test apparatus - The helmet shell assembly shall be tested on a drop type impact machine. (See Figure 1.) The impact shall be supplied

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TABLE I

SAMPLE SIZE, ACCEPTANCE CRITERIA, EXAMINATIONS AND TESTS OF THE HELMET SHELL ASSEMBLIES

INSPECTION	METHOD PARAGRAPH	SAMPLE SIZE	ACCEPTANCE CRITERIA 1/
Visual examination	4.6.1	Every unit for critical defects. Inspection Level II for minor defects.	Reject all units with any critical defects and an Acceptable Quality Level of 15.0 defects per 100 units for minor defects.
Dimensional examination	4.6.1	Inspection Level S-2.	An acceptable quality level of 2.5 defects per 100 units.
Preparation for delivery	4.6.2	Inspection Level S-2.	An acceptable quality level of 2.5 defects per 100 units.

1/ The sampling plan acceptance number shall apply collectively to all the characteristics within a stated acceptable quality level.

TABLE II

CLASSIFICATION OF DEFECTS FOR VISUAL EXAMINATION OF THE HELMET SHELL ASSEMBLY

DEFECT	MINOR	CRITICAL
Shell warped or distorted		X
Gloss of finish less than standard color chip	X	
Orange peel, runs, or sags in paint	X	
Scratched, scuffed, or abraded areas on shell surface	X	
Color not uniform over total surface of shell		X
Label or identification tags missing	X	

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TABLE III

LIST OF DEFECTS FOR FINISHED DIMENSIONS

EXAMINE	DEFECT
Measure all visible components of the helmet shell assemblies.	Any measurement deviating from the dimensions and tolerances as specified in this specification and applicable ACED drawings shall be classified as a defect.

TABLE IV

CLASSIFICATION OF DEFECTS FOR PREPARATION FOR DELIVERY

ITEM	DEFECT
Contents	More than one helmet shell assembly in the same container
Exterior & Interior	Missing, incomplete, incorrect, illegible; of improper size, location, sequence, or method of application; exterior and interior markings different
Packaging and packing material	Inadequate application of the components such as incomplete closure of unit package, intermediate package, case liners, container flaps, loose strappings, etc.; bulging or distortion of containers
Exterior and interior weight or content	Number per container is more or less than required; gross or net weight exceeds the requirement

by dropping a 16.3 pound weight with a 1.9 inch spherical radius impacting surface onto a free swinging head assembly. The head shall be pivoted about an axis 14-1/2" below the apex of the head, and the weight shall be guided by two slide wires running between upper and lower extremities of the machine. The drops shall be run with the perpendicular extremities of the machine. The drops shall be run with the perpendicular axis of the head at that angle (with the horizontal plane) necessary to make the

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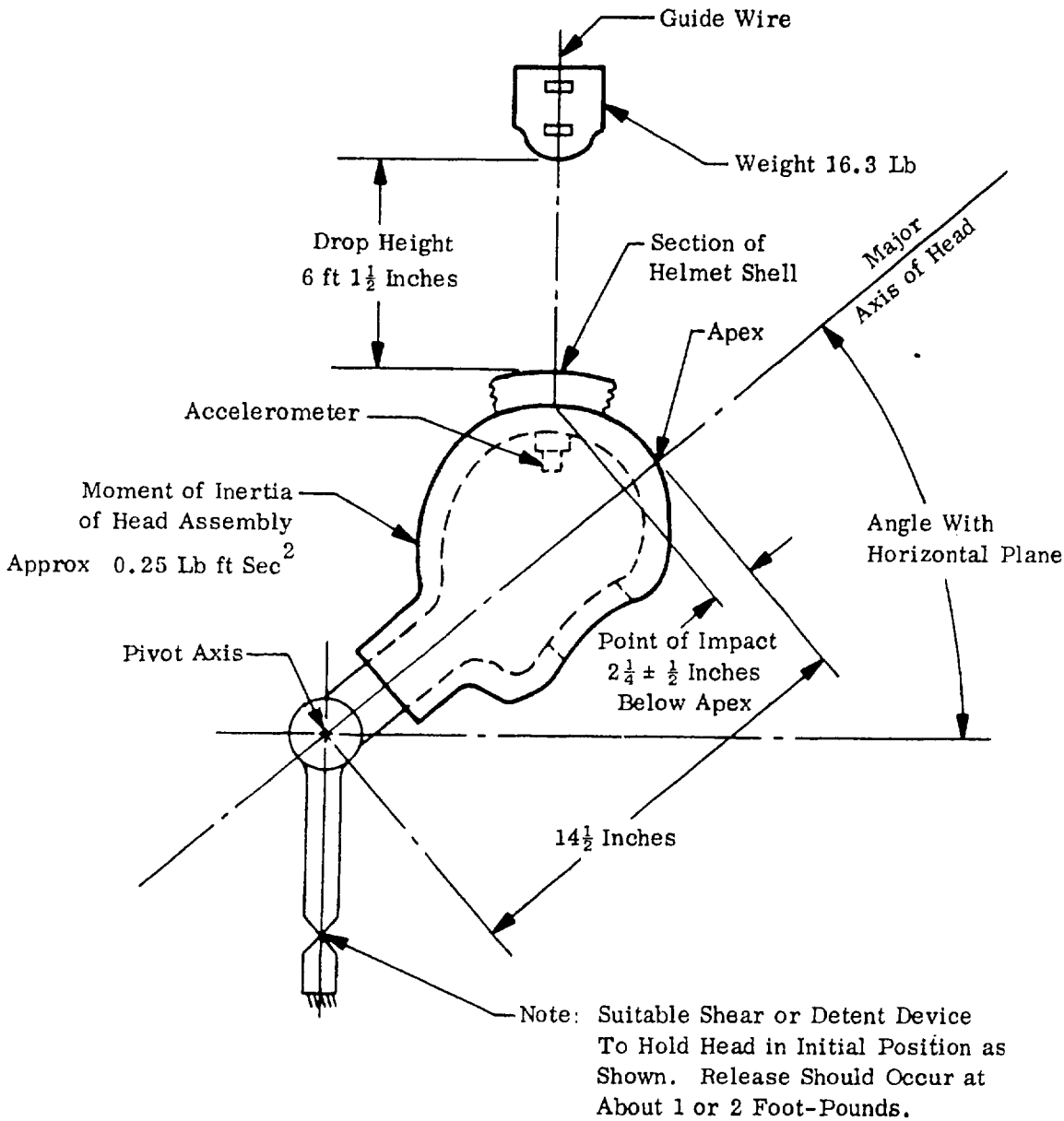


FIGURE 1. Helmet Shell Impact Test Apparatus

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impact normal to the surface of the head. The impact point shall be 2-1/4 ±1/2" from the apex of the head, the distance to be measured along the major axis of the head. A crystal accelerometer shall be hard mounted to the inner surface of the hollow head directly below the point of impact. The accelerometer shall be connected through a 3kc low pass filter to a suitable oscilloscope to record acceleration versus time. The head form shall be of average size so that medium helmet shell assemblies with the thinnest sizing pads and large helmet shell assemblies with the thickest pads provide a reasonable fit. The head shall be cast of high damping magnesium alloy, and shall have an opening approximately 3-1/2" x 3-1/2" in the face area to permit access to the inner surface. The nominal wall thickness of the head shall be about one inch and the casting shall weigh about 13 pounds. The moment of inertia of the pivoted head assembly shall be approximately 0.25 pounds ft. sec^2. A drop height of 6 feet 1-1/2" shall be used. This drop height represents an impact energy of 100 foot pounds.

4.6.4.2 Test - The helmet shell assembly shall be placed on the head form and the weight dropped onto the helmet. The g/time characteristics shall be evaluated from the oscillogram. The helmet shell assemblies shall meet the requirements of 3.5.2. "Bottoming" is indicated by the radical departure of the acceleration versus time curve (g/t) from a smooth energy absorbing curve. "Bottoming" will always cause significant excitation of the natural frequency of the magnesium head form. However, natural frequency excitation of the head can occur without "bottoming" so that the criteria for determining if the inside of the helmet shell assembly has bottomed on the head is the appearance of sharp discontinuities in the rising portion of the g/t curve. See figure 2 for levels on sample helmet shell material. If difficulty is experienced in evaluating impact performance from the oscillograms it will be found helpful to make a second impact in the same spot and compare the second g/t curve with the first. Helmet shell assemblies shall be impacted in four locations, the forehead, both temple, and the occipital regions.

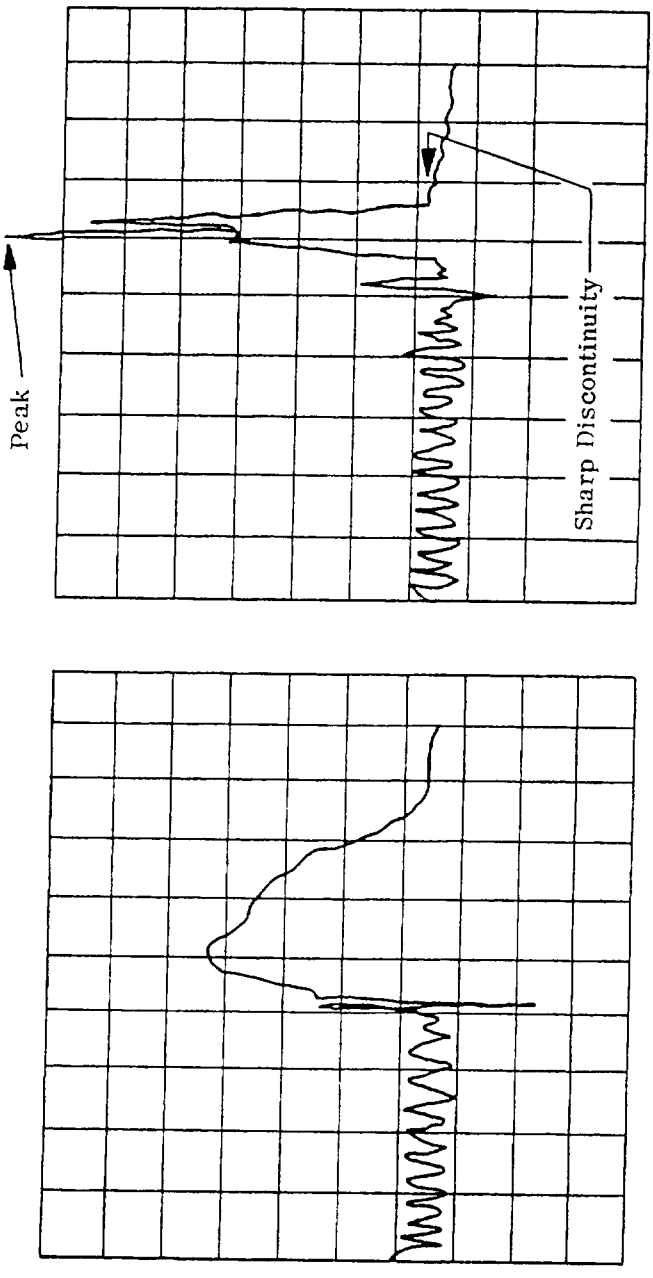
4.6.5 Finish adhesion - With a scribe held at 30° angle, make three parallel, straight lines 1/16 inch apart in any direction; then cross these lines with three additional lines perpendicular and 1/16 inch apart. This procedure produces squares which shall be inspected visually for sharpness of definition and the requirements of 3.5.3. A slight unevenness of the edges of any of the squares shall not be considered cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Application - The packaging, packing and marking requirements specified herein apply only to direct purchases by or direct shipment to the Government.

5.2 Packaging - Levels of packaging shall be as specified by the procuring activity.

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Peak g in Oscillogram No. 2 in Excess of 400 g

FIGURE 2. Sample Impact Oscillogram

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5.2.1 Level A - Unless otherwise specified, each helmet shell assembly with accessories (see 3.4) shall be packed in a unit container in accordance with PPP-B-636, Class Weather Resistant. The container shall be designed to enclose the helmet shell assembly in a snug, glove-like fit to prevent motion of helmet or component parts during shipment, cushioning material used to cushion and prevent such motion shall conform to PPP-C-00795.

5.2.2 Level C - When this level is required, packaging shall be in accordance with standard commercial practice.

5.3 Packing - The helmet shell assemblies packaged as in 5.2.1 shall be of a minimum weight and cube consistent with the specification requirements. Closures of all containers shall be in accordance with the procedures described in the respective container specification or appendices thereto. Unless otherwise specified, the gross weight of the packs shall not exceed approximately 200 pounds.

5.3.1 Level A - The helmet shell assemblies packaged in fiberboard containers as specified in 5.2.1 shall be packed in overseas type exterior containers conforming to PPP-B-591, PPP-B-601, PPP-B-621 or PPP-B-576. Plywood, when used, shall conform to Type I or II, Class 2, of NN-P-530. Each exterior shipping container shall be furnished with case liners, material conforming to UU-P-268, and fabricated in accordance with MIL-L-10547. A case liner for PPP-B-636, Class Weather Resistant containers, is not necessary when all seams and closures are sealed with water resistant tape conforming to PPP-T-60.

5.3.2 Level B - Helmet shell assemblies packaged as in 5.2.1 shall be packed in domestic type exterior containers conforming to PPP-B-591, PPP-B-601, PPP-B-621 or PPP-B-576. Unless otherwise specified, the unit container, with the seams and joints sealed in accordance with the appendix of PPP-B-636, may be used as the shipping container.

5.3.3 Level C - When this level of packing is required, the helmet shell assemblies shall be packed to insure that the shipment arrives satisfactorily at destination. The shipment shall conform to the applicable carriers rules and regulations in effect at the time of shipment.

5.4 Marking - Unit packages, and intermediate containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use - The helmet shell assemblies covered by this specification are intended for use with the APH-6 Series helmet assemblies. The helmet

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assemblies are intended for use by personnel engaged in the operation of high speed aircraft or aircraft undergoing flight test, or both. The helmet assemblies will afford protection against head injury and buffeting.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Quantity and sizes of helmet shells desired

(c) Whether first article inspection is waived (see 4.4.1)

(d) Name and address of first article inspection laboratory (see 4.4.1)

(e) Selection of applicable levels of packaging and packing (see 5.1, 5.2 and 5.3)

6.3 Qualification - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command, Department of the Navy, Washington, D.C. 20360; however, authorization for qualification of products shall be obtained from the Commanding Officer, Naval Air Development Center, Johnsville, Warminister, Pennsylvania 18974, Attention: Director, Aerospace Crew Equipment Department. Prior to submission of the samples for qualification inspection, the manufacturer shall submit a request to the Aerospace Crew Equipment Department indicating a date on which the samples can be forwarded and also request an authorization number to accompany the samples.

6.3.1 When requested, the manufacturer will submit engineering drawings and inspection reports in accordance with SD-6, Provisions Governing Qualification.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-H-22995A(AS) HELMET SHELL ASSEMBLIES, PILOT'S PROTECTIVE, APH-6 SERIES

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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Naval Air Systems Command
Washington, D.C. 20360

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